

Understanding Support Vector Machines and the Impact of Kernel Functions

1. Introduction

Machine learning is widely used for analysing numerical datasets and discovering patterns that traditional statistical approaches may fail to detect. Among supervised learning algorithms, Support Vector Machines (SVM) remain one of the most powerful tools for classification tasks. The algorithm constructs a decision boundary that separates classes while maximising the margin between them. However, many real-world datasets are not linearly separable. SVM addresses this limitation using kernel functions which map data into higher dimensional feature spaces. This tutorial demonstrates how different kernel functions influence model behaviour and performance using the Breast Cancer Wisconsin dataset.

2. Support Vector Machine Fundamentals

Support Vector Machines attempt to identify the optimal hyperplane that separates two classes. The optimal hyperplane maximises the margin between the decision boundary and the closest points from each class. The closest points are known as support vectors and determine the position of the hyperplane. Because only these points influence the model, SVMs are often robust to noise and effective in high dimensional spaces.

3. The Kernel Trick

In nonlinear datasets, a straight decision boundary cannot separate the classes. The kernel trick enables SVM to transform input data into higher dimensional spaces where linear separation becomes possible. Kernel functions compute similarity between data points without explicitly calculating the transformation. This approach makes it computationally efficient to model complex patterns.

4. Common Kernel Functions

The most common kernels used with SVM include:

- Linear Kernel – works well when the dataset is approximately linearly separable.
- Polynomial Kernel – models interactions between features and produces curved boundaries.
- Radial Basis Function (RBF) Kernel – highly flexible and capable of capturing complex nonlinear patterns.
- Sigmoid Kernel – inspired by neural network activation functions.

5. Dataset Description

This tutorial uses the Breast Cancer Wisconsin dataset available in the Scikit-learn library. The dataset contains 569 observations and 30 numerical features describing cell nuclei characteristics. The task is binary classification: determining whether a tumour is benign or malignant.

6. Methodology

The experiment follows a standard machine learning workflow.

1. Load dataset using Scikit-learn
2. Split data into training and testing sets
3. Apply feature scaling using standardisation
4. Train SVM

models with different kernels 5. Evaluate performance using accuracy and confusion matrices 6. Visualise decision boundaries

7. Results and Analysis

Results show that kernel choice significantly influences classification performance. The RBF kernel often provides the highest accuracy because it adapts well to nonlinear structures in the data. Linear kernels perform well when the dataset is nearly linearly separable, while polynomial kernels provide additional flexibility. Decision boundary visualisations demonstrate how different kernels produce different classification regions.

8. Ethical Considerations

Machine learning systems are increasingly used in healthcare environments such as cancer diagnosis. However, ethical concerns must be considered. Dataset bias may cause models to perform poorly on underrepresented populations. False predictions may lead to unnecessary medical procedures or delayed treatment. Therefore machine learning systems should support rather than replace medical professionals.

9. Conclusion

Support Vector Machines remain a powerful algorithm for classification tasks. Understanding kernel functions is essential for selecting appropriate models for real-world machine learning problems. The comparison presented in this tutorial highlights how kernel selection affects both performance and decision boundary structure.

References

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